

## Prewarming

Warm chosen nodes *before* jobs arrive.

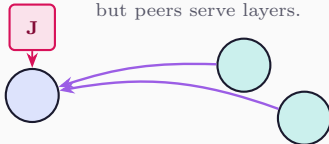


**fewer cold hits:**  $c/N \downarrow$

Changes *whether* a job hits a cold node.  
Reduces the number of affected jobs.

## Cluster-local mirror

A cold hit still happens,  
but peers serve layers.



**cheaper pulls:**  $p_I \downarrow$

Changes *how costly* a cold hit is. Re-  
duces  $p_I$ , not the hit probability.

**Complementary:** prewarming avoids cold hits; a mirror makes the remaining cold hits cheaper.